

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

SGCC XGBoost Hyperparameter Tuning & Threshold Optimization Report

1. Tuning Methodology

- **Search Space:** `n_estimators`, `max_depth`, `learning_rate`, `min_child_weight`, `subsample`, `colsample_bytree`, `gamma`, `reg_alpha`, `reg_lambda`
- **CV Strategy:** 5-fold StratifiedKFold on the 29,656-row Training split (Random Seed 42). 20 randomized iterations.
- **Data Scaling/Imputation:** `SimpleImputer(strategy='median')` applied internally within each CV fold to strictly prevent leakage.
- **Optimization Metric:** Macro F1 (refit).

2. Selected Hyperparameters

`{'xgb__subsample': 0.7, 'xgb__reg_lambda': 100, 'xgb__reg_alpha': 1, 'xgb__n_estimators': 200, 'xgb__min_child_weight': 10}`

3. CV & Validation Performance

- **Cross-Validation Mean F1:** 0.409
- **Validation F1 (0.50 Threshold):** 0.457

4. Threshold Analysis

- Evaluated threshold grid: `[0.05, 0.95]` strictly on Validation predictions.
- **Optimal F1 Threshold:** 0.50
- The validation F1 score at this optimal threshold is **0.457**.
- A stable threshold region was observed in the precision-recall plots (see `graphs/xgboost_tuning/`).

5. Final Sealed Test Evaluation

Configuration	Threshold	Precision	Recall	F1	PR-AUC	ROC-AUC
Baseline XGB (Weighted)	0.50	0.353	0.411	0.380	0.344	0.803
Tuned XGB	0.50	0.345	0.483	0.402	0.375	0.826
Tuned XGB (Optimal)	0.50	0.345	0.483	0.402	0.375	0.826

6. Interpretation

- **Baseline vs Tuned:** Hyperparameter tuning slightly shifted the model's capacity.
- **Validation-to-Test Gap:** The test performance is closely aligned with validation, indicating no severe overfitting.
- **Precision/Recall Tradeoff:** Operating at the optimal threshold balances identifying theft cases (Recall) against overwhelming investigators with false positives (Precision).
- **Note:** Missingness features demonstrated predictive association in the baseline phase; this tuning phase incorporates them but does not assert causality.

7. Limitations

- We have not yet pruned correlated features (Feature Selection), which tree models can sometimes struggle with when highly collinear variables exist.